

WEIKAI LIAO

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EDUCATION

Duke University

Master's, Electrical and Computer Engineering

May 2027

University of Electronic Science and Technology of China

Bachelor's, Software Engineering

June 2024

PROFESSIONAL EXPERIENCE

Microsoft

Research Intern

March 2026 - July 2026

- Architected HDMAS, a planner-free, decentralized multi-agent architecture using a lock-protected shared blackboard and distributed task claiming. Holding the GPT-5.4 Copilot backend constant, the multi-agent system improved Row-F1 by **11.4%** and Item-F1 by **8.0%** over a single agent on 200 WideSearch tasks.
- Built an automated three-stage benchmark generation pipeline that decomposes tasks into atomic facts, generates ground truth from the live web, and independently verifies each fact; increased verified ground-truth coverage from **22.7% to 70.1%**.
- Post-trained a batched code-act web-search agent on Qwen2.5-7B (LoRA SFT + GRPO): SFT improved F1 **8×** (5% to 40%); a custom-reward RL pass added **+9%** F1 while cutting per-query cost **14%** and latency **9%**.

Alibaba Cloud

Software Development Engineer Intern

July 2025 - November 2025

- Built an A2A protocol adapter for AgentScope Runtime, part of a **27K+**-star open-source agent framework, enabling cross-framework agent interoperability through bidirectional JSON-RPC translation and asynchronous FastAPI streaming.
- Built AgentScope Runtime deployment tooling, automating packaging, OSS uploads, and serverless deployment on Bailian for one-click releases and versioning.
- Developed Bailian's full-code agent management backend using Spring Boot, implementing search/filtering, SLR permission validation, lifecycle management, and database CRUD with generated Mappers and indexes; unified visibility into Function Compute deployment metadata and runtime status across **10K+** agent applications.

BoulderAI Technologies

Software Engineer Intern

April 2024 - June 2024

- Built LLM-driven enterprise automation pipelines using FastGPT, orchestrating prompt workflows to translate natural language into BPMN diagrams and reduce manual process modeling effort by **40%**.
- Built a RAG-powered Excel data extraction tool to identify and structure logical entities from headers, using LLMs to generate and classify structured JSON outputs for data integration.

PROJECTS & OUTSIDE EXPERIENCE

BrowserUse Bot

Team Leader

March 2026 - May 2026

- Built a supervised browser-automation agent controlled via Telegram, with a noVNC-based human-in-the-loop login handoff for authenticated sessions.
- Extended the open-source browser-use agent with a CodeAct tool layer and reusable skills, cutting LLM round-trips by **~95%** and increasing task yield from 1 to 14 grounded results.
- Instrumented the LLM pipeline with Langfuse tracing (per-step tokens, latency, cost, errors), enabling request-ID 'trace lookup to debug production runs and attribute per-task cost.
- Built a GitHub Actions CI/CD pipeline auto-deploying to AWS EC2 with health-check rollback, running the agent 24/7; reduced deploys to a single push (from manual multi-step to ~2 min, zero-downtime).

SKILLS

Programming Language: Java, C/C++, Python, JavaScript, SQL

Technologies: Spring Boot, Django, FastAPI, Vue.js, React, MySQL, PostgreSQL, Redis, Elasticsearch, Docker, AWS, LLM Agents, RAG, SFT, Langfuse, GRPO